

Feed back Control → using information about deviation of the system from its desired state to control the system

Closed loop
feed back
(Automatic)

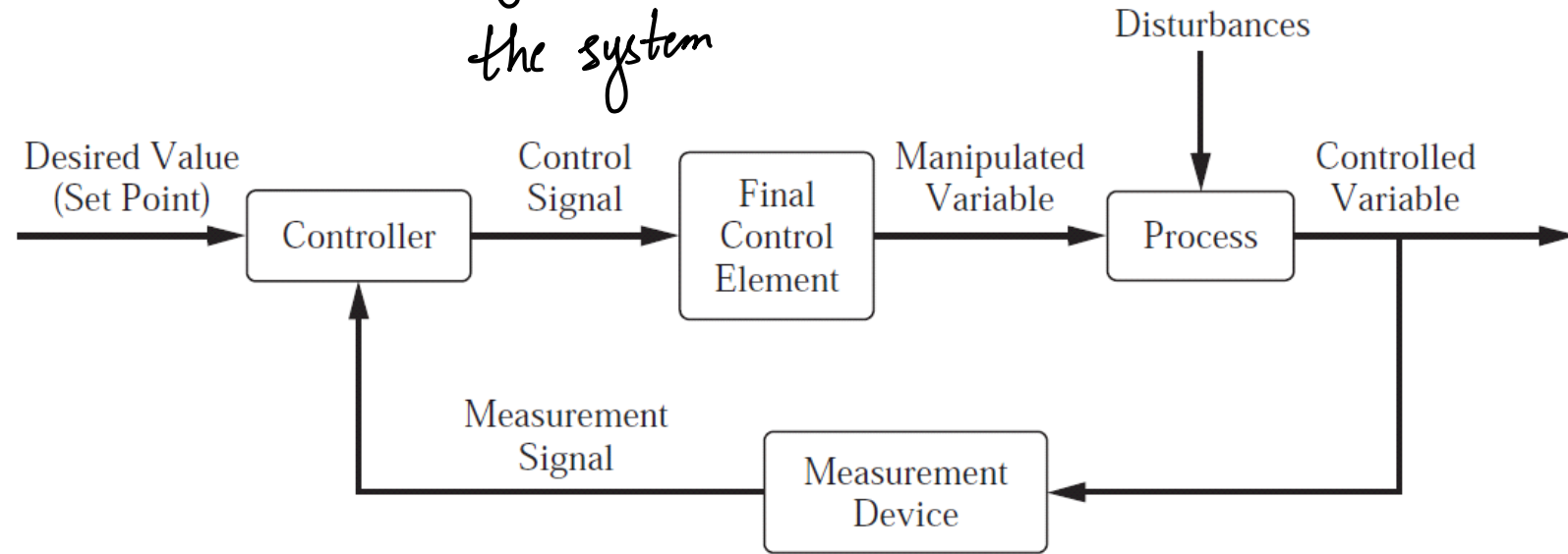


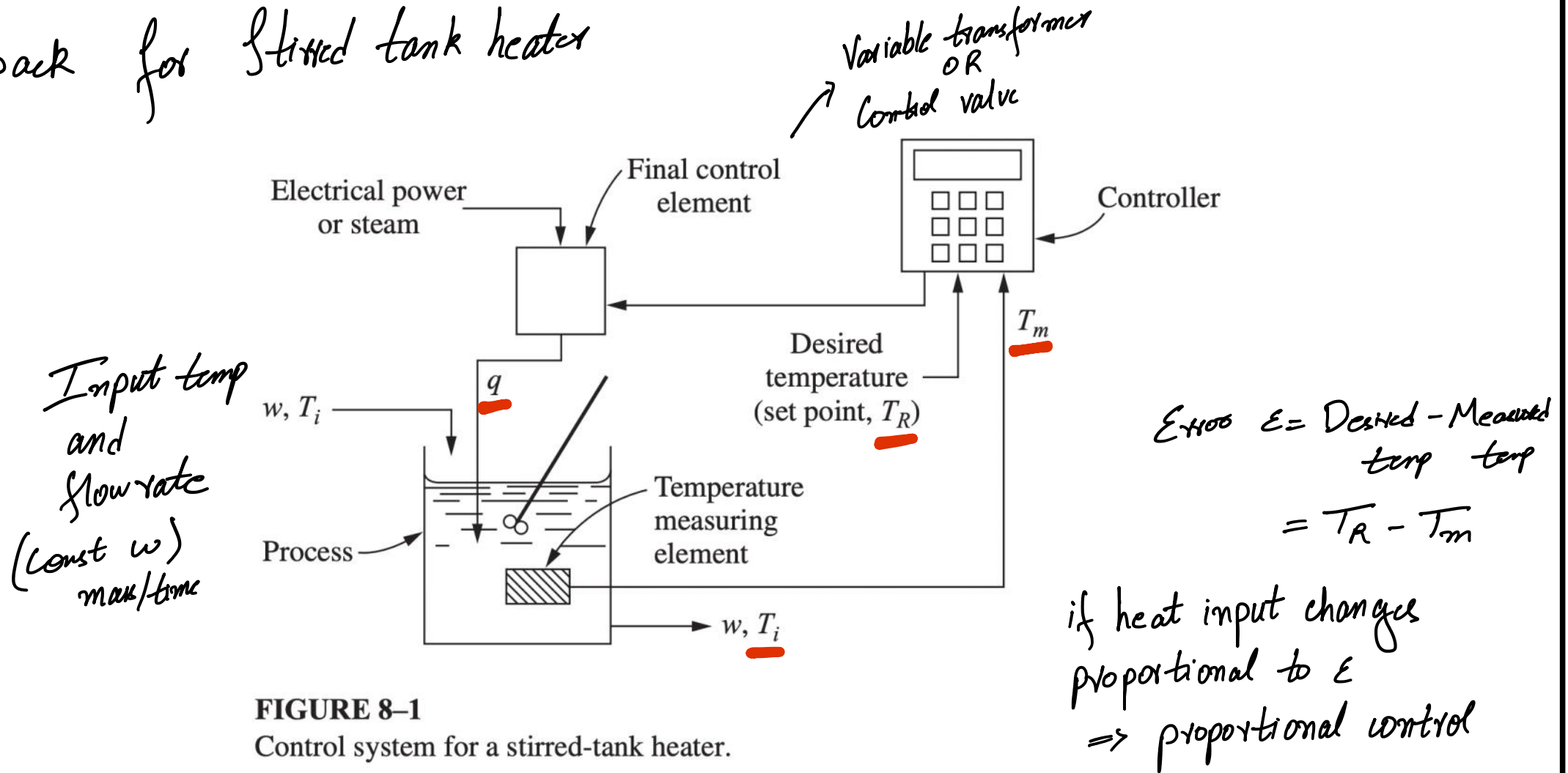
FIGURE 1-1
Generalized process control system.

Open loop system (Manual)

- ↳ Measurement signal disconnected from controller
- ↳ Controller output manually adjusted to change controlled variable

The Control System

→ Feedback for stirred tank heater



Components of a Control System

- 1) Process (Stirred Tank Heater)
- 2) Measuring Element (Thermometer)
- 3) Controller
- 4) Final Control Element

Load $\rightarrow$ Change in any variable that may cause the "control variable" of the process to change
e.g. flow rates, heat losses etc

Block
Diagram

Closed loop/Feedback
control system

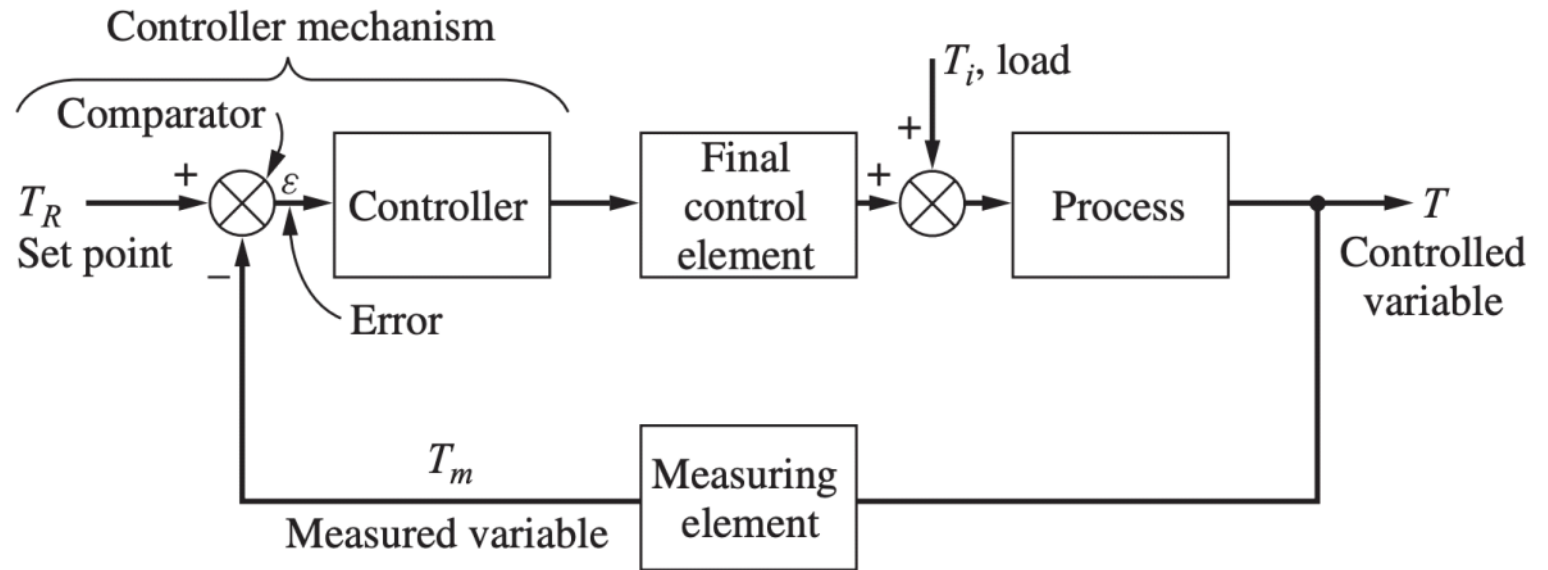


FIGURE 8-2

Block diagram of a simple control system.

Negative Feedback VS Positive Feedback

→ -ive Feedback uses diff $T_R - T_m \rightarrow$ Reduces Error ϵ

→ @ S.S $T = T_R = T_m$

with disturbance of Load T_i

where $T_i \uparrow \Rightarrow T_m \uparrow \Rightarrow \epsilon$ becomes -ive $\Rightarrow q \downarrow$

→ +ive Feedback uses sum $T_R + T_m \rightarrow$ Inherently unstable

→ if $T_i \uparrow \Rightarrow T_m \uparrow \Rightarrow \epsilon \uparrow \Rightarrow q \uparrow$

→ Causes runaway T

- Self heating reactors
- Auto catalytic Reactions

Servo mechanism - type Problem (Set point changes)

↳ No Load changes to account for

↳ T_R may be changed as desired

→ Predetermined time dependant pattern of change

↳ Following a moving set point

Regulator Problem (Load changes)

↳ Keep T_R fixed as T_i changes

Development of Block Diagram

- Uses deviation variables
- Blocks contain transfer functions
- Blocks are combined for overall block diagram

For Stirred Heating tank

→ Unsteady State Energy Balance

$$wC(T_i - T_o) - wC(T - T_o) + q_f = \rho CV \frac{dT}{dt}$$

→ @ S.S

$$wC(T_{is} - T_o) - wC(T_s - T_o) + q_s = 0$$

→ Deviation variables

$$wC(T_i' - T') + Q = \rho CV \frac{dT'}{dt}$$

* q_f in case for using steam is $f(H_g - H_l)$

flow rate Enthalpy gas Enthalpy liq

→ Laplace

$$wC [T_i'(s) - T'(s)] + Q(s) = \rho CV s T'(s)$$

$$wC T'(s) + \rho CV s T'(s) = wC T_i'(s) + Q(s)$$

$$T'(s) [wC + \rho CV s] = wC T_i'(s) + Q(s)$$

$$T'(s) = \frac{wC}{wC + \rho CV s} T_i'(s) + \frac{1}{wC + \rho CV s} Q(s)$$

$$= \frac{1}{1 + \frac{\rho V}{w} s} T_i'(s) + \frac{1/wC}{1 + \frac{\rho V}{w} s} Q(s)$$

1st order
Heater
Transfer function

$$T'(s) = \frac{1}{1 + \tau s} T_i'(s) + \frac{1/wC}{1 + \tau s} Q(s)$$

Outlet Temp
Inlet temp
Heater input

$$\Rightarrow \tau = \frac{\rho V}{w} [=] \text{ sec}$$

$$\Rightarrow \text{Gain for } Q(s) = \frac{1}{wC} [=] \frac{1}{\frac{\text{kg}}{\text{min}} \frac{\text{kJ}}{\text{kg}^\circ\text{C}}} [=] \frac{^\circ\text{C}}{\text{kJ/min}}$$

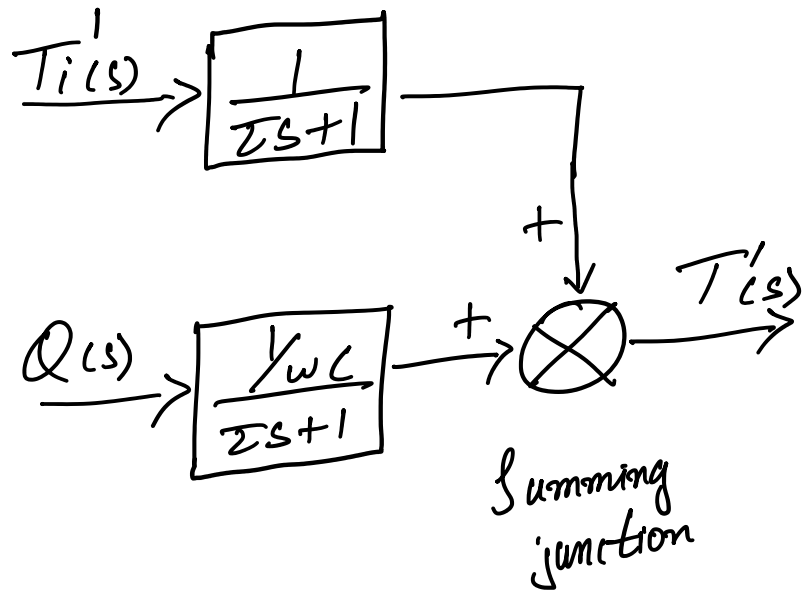
Effect of Heat input and Inlet temp are additive

if only $Q(t)$ changes $\Rightarrow T_i'(t) = 0$

$$\Rightarrow \frac{T'(s)}{Q(s)} = \frac{1/wC}{\tau s + 1}$$

if only $T_i(t)$ changes $\Rightarrow Q(t) = 0$

$$\Rightarrow \frac{T'(s)}{T_i'(s)} = \frac{1}{\tau s + 1}$$

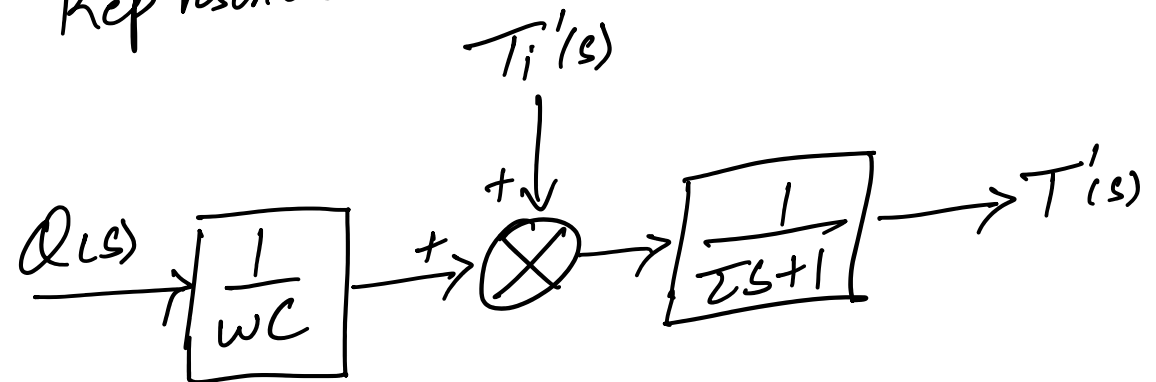


Similarly

$$T'(s) = \frac{1/wC}{\tau s + 1} Q(s) + \frac{1}{\tau s + 1} T_i'(s)$$

$$T'(s) = \left[\frac{1}{wC} Q(s) + T_i'(s) \right] \frac{1}{\tau s + 1}$$

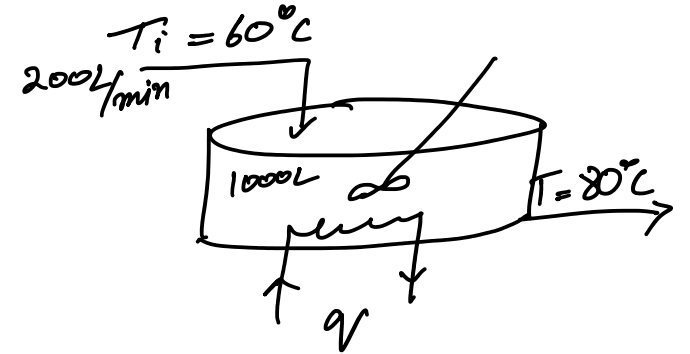
Represented as



Example 8.1

Stirred tank heater

- a) Determine response of outlet temp of tank to a step change in the inlet temp from 60 to 70°C



Starting with

$$T'(s) = \frac{1}{\tau s + 1} Q(s) + \frac{1}{\tau s + 1} T_i'(s)$$

Time const

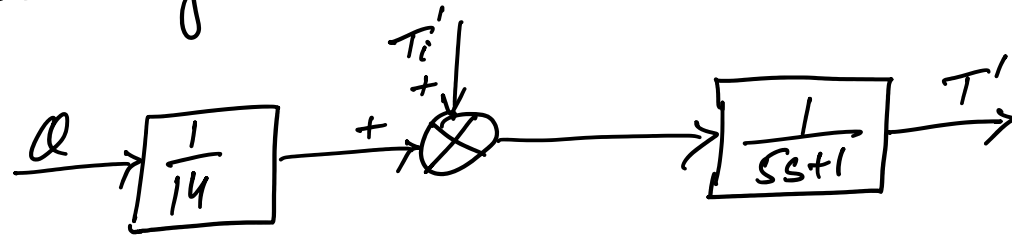
$$\tau = \frac{SV}{w} = \frac{V}{w/\rho} = \frac{V}{v} = \frac{1000L}{200L/min} = 5 \text{ min}$$

Gain

$$\frac{1}{wC} = \frac{1}{SV C} = \frac{1}{\left(\frac{1 \text{ Kg}}{L} \times \frac{200L}{min} \right) \times \frac{1 \text{ min}}{60 \text{ sec}} \times \frac{4.184 \text{ KJ}}{\text{Kg } ^\circ\text{C}}} = \frac{1}{14} \frac{^\circ\text{C}}{\text{KW}} = \frac{1}{14} \frac{^\circ\text{C}}{\text{KW}}$$

$$T'(s) = \left(\frac{1}{14} \frac{^\circ\text{C}}{\text{KW}} \right) \frac{1}{5s + 1} Q(s) + \frac{1}{5s + 1} T_i'(s)$$

Block diagram



for q_s using S.B

$$wC(T_{is} - T_o) - wC(T_s - T_o) + q_s = 0$$

$$q_s = wC(T_s - T_{is})$$

$$= 200 \frac{\text{Kg}}{\text{min}} \frac{1 \text{ min}}{60 \text{ sec}} \cdot 4.184 \frac{\text{KJ}}{\text{Kg}^\circ\text{C}} (80 - 60) = 280 \text{ KW}$$

For Step change $60 - 70^\circ\text{C}$, $T_i'(t) = 70 - 60 = 10^\circ\text{C} \Rightarrow T_i'(s) = \frac{10}{s}$
 with $Q(s) = 0$

$$T'(s) = \frac{10}{s} \frac{1}{5s+1}$$

$$T'(t) = 10(1 - e^{-t/5}) \Rightarrow T(t) = T_s + T'(t) = 80 + 10(1 - e^{-t/5})$$

$$T' = T - 80$$

$$T_i' = T_i - 60$$

$$Q = q' - q_s \\ = q - 280 \text{ KW}$$

b) Determine the response of the outlet temperature of tank to a step increase in the heat input of 42 kW

$$T_i' = 0$$

$$T'(s) = \frac{42}{s} \cdot \frac{1}{14} \cdot \frac{1}{s+1}$$

$$T'(t) = 3(1 - e^{-t/5})$$

$$\Rightarrow T(t) = T_s + T'(t) = 80 + 3(1 - e^{-t/5})$$

↑
ultimate change in
temp is 3°C

c) Determine the response of the outlet temp of tank to simultaneous step change in inlet temp from 60 to 70°C and heat input of 42 KW

$$T'(s) = \frac{42}{s} \frac{1}{14} \frac{1}{5s+1} + \frac{10}{s} \frac{1}{5s+1}$$

$$= 3(1 - e^{-t/5}) + 10(1 - e^{-t/5})$$

$$= (1 - e^{-t/5})(3 + 10)$$

$$= 13(1 - e^{-t/5})$$

Measuring Element

↳ lag in temp detection $\Rightarrow$ first order lag seen for mercury thermometer

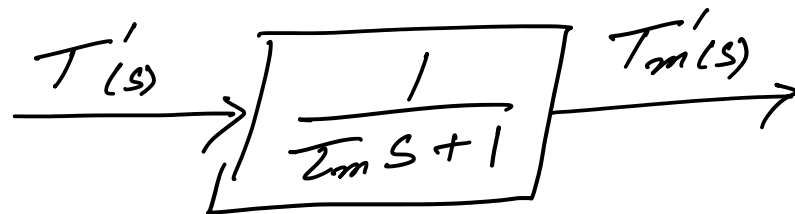
$$\text{Transfer function} = \boxed{\frac{T_m'(s)}{T'(s)} = \frac{1}{\tau_m s + 1}}$$

$$T' = T - T_s$$

$$T_m' = T_m - T_{ms}$$

$\tau_m \sim 6 - 20 \text{ sec}$
for thermocouples

Block Diagram



For a unit step change in temp $T'(s) = \frac{1}{s}$

Response is :-

$$T_m'(s) = \frac{1}{s} \frac{1}{\tau_m s + 1}$$

$$\Rightarrow T_m'(t) = (1 - e^{-t/\tau_m})$$

$$\Rightarrow \frac{T_m'(s)}{T'(s)} = \frac{1}{0.33s + 1}$$

Example 8.2

Response time = 45 sec $\rightarrow$ Reading is 90% complete

$$\Rightarrow 0.9 = 1 - e^{-45/\tau_m}$$

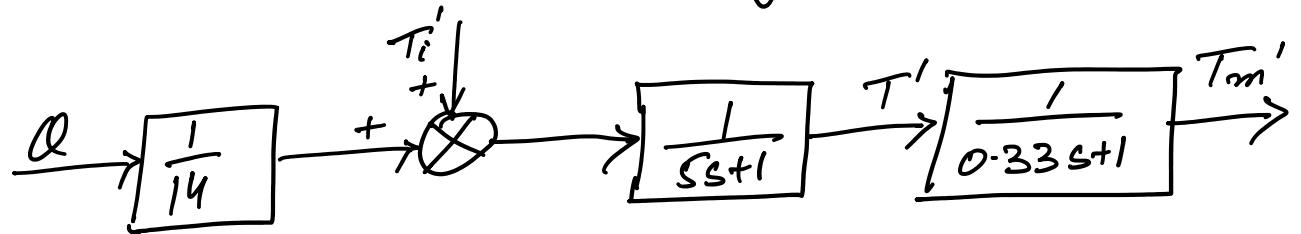
$$1 - 0.9 = e^{-45/\tau_m}$$

$$\ln(0.1) = \frac{-45}{\tau_m}$$

$$\tau_m = \frac{-45}{\ln(0.1)} = \frac{-45}{-2.302} = 19.54 \text{ sec}$$

$$\tau_m = 0.33 \text{ min}$$

Block diagram



Controller and Final Control Element

↳ Assuming a Proportional controller

↳ One block representing controller and final control element

For a proportional controller:-

$$q = K_c E + A$$

↓
Proportional
sensitivity
or Gain

↓
Heat input
when $E=0$
 $A = q/s$

$$; E = T_R - T_M$$

↓
Setpoint

At Steady State

$$T_{R_s} = T_s = T_{m_s}$$

$E' \rightarrow$ Deviation variable

$$\Rightarrow @ \text{S.S } E' = E - 0 = E$$

$$E' = E - E_s ; E_s = T_{R_s} - T_{m_s} \Rightarrow @ \text{S.S } E_s = 0 \text{ as } T_{R_s} = T_{m_s}$$

@ S.S there is no error

$$q' = K_c E + A$$

$$q'_s = K_c \cancel{E}_s + A$$

$$q'_s = A$$

q'_s is required to raise temp from T_{is} to T_{Rs} called bias value

$$q'_s = wC (T_{Rs} - T_{is})$$

$$\Rightarrow q' = K_c E + q'_s$$

$$q' - q'_s = K_c E$$

$$Q = K_c E$$

Laplace

$$Q(s) = K_c E(s)$$

Proportional controller transfer function

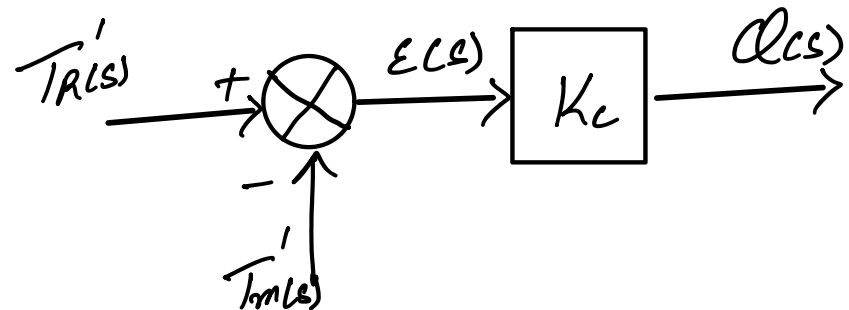
$$E = (T_R - T_{Rs}) - (T_m - T_{ms})$$

$$E = T_R' - T_m'$$

$$E(s) = T_R'(s) - T_m'(s)$$

Error definition for stirred tank heater

Block Diagram



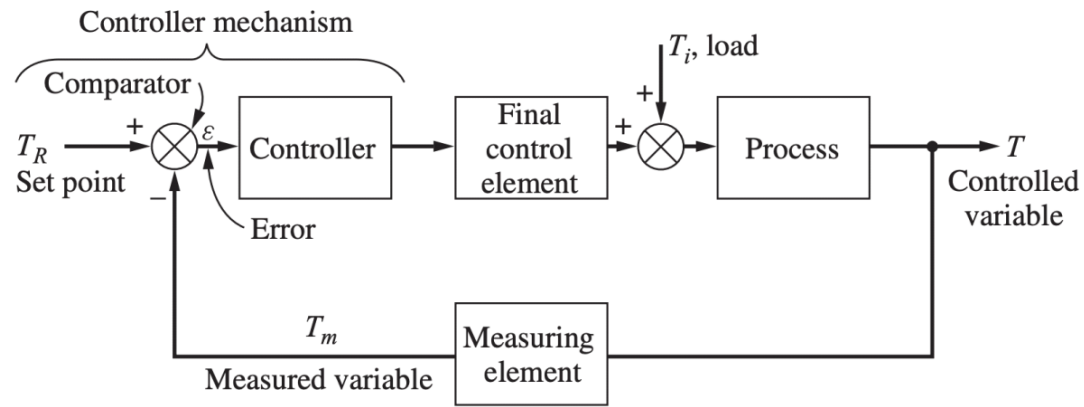


FIGURE 8-2
Block diagram of a simple control system.

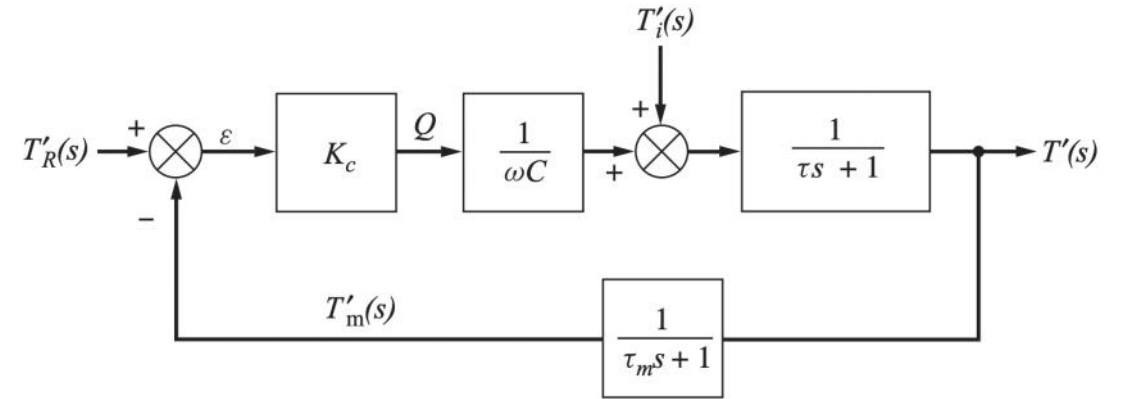


FIGURE 8-16
Block diagram of control system.

